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ITAI 4370: AI 5/6G Comm & ORAN Net

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Assignment 4

Section A: Conceptual Understanding

Question 1 - *What is Ethical AI, and why is it essential in modern technological systems?*

Ethical AI means designing and using AI in a way that respects people, their rights, and society, rather than only chasing performance. It covers fairness, transparency, accountability, privacy, and the avoidance of harm (Slide 4, Slide 12). It is essential because AI now makes or shapes decisions that affect real lives, in areas like hiring, lending, policing, and healthcare, so a biased or opaque system can cause harm at scale and wear away public trust (Slide 4). Building ethics in from the start helps prevent that harm and keeps AI aligned with human values.

Question 2 - *List and explain any four major ethical principles that guide AI development.*

Four of the core principles that guide AI development (Slide 12) are:

- **Fairness:** AI should treat people equally and avoid discriminating against any group.
- **Transparency:** how a system reaches its decisions should be open and understandable, not a black box.
- **Accountability:** there must be a clear answer to who is responsible when an AI system causes harm.
- **Privacy and autonomy:** people's data should be protected and their right to make their own choices respected.

Question 3 - *What is the main objective of the IEEE Ethically Aligned Design framework?*

The IEEE Ethically Aligned Design framework, whose full title is A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, has one central aim: to put human well-being first in the design of AI and autonomous systems (IEEE Standards Association, 2019). Rather than measuring success only by technical performance, it asks designers to make sure systems are beneficial to people and the environment, backed by guidelines and standards that keep them aligned with human values (IEEE Standards Association, 2019).

Question 4 - *Define the term algorithmic bias and provide one real-world example.*

Algorithmic bias is when an AI system produces unfair or skewed results that favor or disadvantage certain groups, usually because of biased training data, poor labeling, or flawed design (Slide 6). A real-world example is facial recognition that misidentifies people of color far more often than white faces, because the data used to train it underrepresented darker-skinned people (Slide 6).

Question 5 - *Explain the difference between transparency and explainability in AI.*

Transparency is about openness: being clear about how a system works, what data it uses, and how it was built. Explainability is narrower: it is the ability to explain, in terms a person can understand, why the system reached a particular decision (Slide 8). A model can be transparent about its design and still be hard to explain if it is a complex black box, which is why explainable AI focuses on making individual decisions clear enough for users to trust (Slide 8).

Question 6 - *What role does accountability play in the ethical use of AI systems?*

Accountability answers the question of who is responsible when an AI system fails or causes harm (Slide 9). It matters because AI often sits between several parties, the developer, the company that deploys it, and the user, so without clear accountability harm can happen with no one answerable. It is supported by practices like keeping a human in the loop and running ethical audits, so decisions can be traced, reviewed, and corrected (Slide 9).

Question 7 - *Identify three international organizations involved in developing AI ethics guidelines.*

Three international organizations involved in developing AI ethics guidelines are (Slide 11):

- **OECD:** its AI Principles were the first intergovernmental standard on AI (OECD, 2024).
- **UNESCO:** it produced the first global standard on AI ethics in 2021 (UNESCO, 2021).
- **The European Union:** it turned these ideas into law with the EU AI Act (Slide 22).

Question 8 - *What does UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasize?*

UNESCO's Recommendation, adopted in 2021 as the first global standard on AI ethics, puts the protection of human rights and human dignity at its center (UNESCO, 2021). It is built on principles like transparency, fairness, environmental sustainability, and keeping humans in oversight of AI, and it calls on governments to set up real laws and institutions rather than leaving AI to regulate itself (UNESCO, 2021).

Question 9 - *What are OECD's five key principles for trustworthy AI?*

The OECD's five values-based principles for trustworthy AI are (OECD, 2024):

- Inclusive growth, sustainable development, and well-being.
- Respect for human rights, democratic values, and fairness.
- Transparency and explainability.
- Robustness, security, and safety.
- Accountability.

Question 10 - *How does the EU AI Act classify and regulate AI systems based on risk?*

The EU AI Act uses a risk-based approach and sorts AI systems into four levels (Future of Life Institute, 2024):

- **Unacceptable risk:** banned outright, such as social scoring and manipulative systems.
- **High risk:** allowed but tightly regulated, with requirements like risk management, documentation, and human oversight, for uses such as hiring, credit scoring, and law enforcement.
- **Limited risk:** allowed with transparency duties, such as telling users they are talking to a chatbot or that content is a deepfake.
- **Minimal risk:** everything else, such as spam filters, with no mandatory obligations.

The higher the risk to people's rights and safety, the stricter the rules (Future of Life Institute, 2024).

Section B: Applied and Scenario-Based

Question 11 - *An AI recruitment tool consistently prefers male candidates. Which ethical principles are violated?*

This mainly violates fairness and non-discrimination, because the tool treats candidates unequally based on gender (Slide 12). It also breaks accountability, since someone deployed a biased system and is answerable for the harm, and it touches transparency, because the bias likely went unnoticed inside an unexamined black box (Slide 9, Slide 12). The root cause is usually biased training data, for example resumes from a past workforce that was mostly male.

Question 12 - *A self-driving car must choose between hitting one pedestrian or five. Which ethical dilemma does this represent?*

This is the classic trolley problem, applied to autonomous vehicles (Slide 18). It forces a choice about the value of human life and whether a machine should be programmed to minimize deaths even if that means actively harming someone (Slide 18). It has no clean answer, which is why it drives calls for shared, agreed ethical standards for how self-driving cars should behave in unavoidable accidents (Slide 18).

Question 13 - *In predictive policing, how can bias be introduced into the algorithm's training data?*

Predictive policing learns from historical crime data, and if past policing was biased, the data carries that bias forward (Slide 6, Slide 16). If certain neighborhoods were over-policed, they show more recorded arrests, so the model sends more patrols there, which produces still more arrests and a feedback loop that looks objective but simply repeats old patterns (Slide 16). Skewed or poorly labeled data is the main entry point for this bias (Slide 6).

Question 14 - *How might a "human-in-the-loop" system reduce ethical risks in automated decision-making?*

A human-in-the-loop system keeps a person involved in reviewing or approving the AI's decisions rather than letting it act fully on its own (Slide 9). This adds a checkpoint where a human can catch errors, question unfair outcomes, and take responsibility, which reduces the risk of harmful automated decisions going unchecked (Slide 9). It is especially valuable in high-stakes areas like healthcare, lending, and criminal justice.

Question 15 - *Describe how data privacy laws such as GDPR affect the design of AI systems.*

Privacy laws like the GDPR force privacy to be built into an AI system from the start rather than added later (Slide 7). In practice this means collecting only the data actually needed, having a legal reason or consent to use it, being clear about how it is used, and letting people access or delete their data under the right to be forgotten (Slide 7). The GDPR also gives people protection against purely automated decisions, so designers often keep a human in the loop and run impact assessments before deploying AI that uses personal data.

Question 16 - *A generative AI creates a fake video of a politician. What ethical issues arise here?*

This raises the problem of deepfakes and misinformation: a convincing fake can spread lies, damage a person's reputation, and mislead voters (Slide 19). It also raises questions of authenticity and consent, since the politician's likeness is used without permission, and of accountability, since it can be hard to trace who made it (Slide 19). It is a key reason the field is pushing for ways to verify authentic content and to label AI-generated media (Slide 19).

Question 17 - *What are the ethical implications of AI-powered surveillance systems in public spaces?*

Public AI surveillance, such as facial recognition, collects data on people who never agreed to it, which threatens privacy and the freedom to move and gather without being tracked (Slide 7). It can enable mass monitoring and state or corporate control, and it can be biased, misidentifying some groups more than others and leading to wrongful targeting (Slide 6, Slide 7). The core tension is between security and individual rights, which is why the EU AI Act tightly restricts real-time biometric identification in public (Future of Life Institute, 2024).

Question 18 - *How can organizations ensure transparency in AI-driven financial decisions?*

They can use models whose decisions can be explained, so a customer denied a loan can be told the actual reasons rather than being left with a machine saying no (Slide 8). Supporting steps include documenting how the system works, disclosing when AI is being used, auditing it for bias, and keeping a human available to review contested decisions (Slide 8, Slide 9). This kind of explainability is what builds trust in automated financial decisions (Slide 8).

Question 19 - *Discuss the ethical concerns surrounding AI in healthcare diagnostics.*

In healthcare, a wrong AI diagnosis can directly harm a patient, so accuracy and safety are the first concern (Slide 5). Bias is another: if the training data underrepresented some groups, the tool may work worse for them and widen health gaps (Slide 6). There are also questions of accountability, who answers if the AI is wrong, the doctor or the developer, and of privacy, since medical data is highly sensitive, which is why a human clinician should stay in the loop rather than hand decisions fully to the machine (Slide 9).

Question 20 - *What measures can be implemented to make AI datasets more representative and fair?*

The main measure is to use diverse, well-balanced data that includes the groups the system will affect, so no group is underrepresented (Slide 6). This is backed by careful, accurate labeling, checking data for gaps and skew, running fairness audits before and after deployment, and updating the data as problems appear (Slide 6). The goal is to catch and correct bias in the data before it becomes bias in the decisions.

Section C: Reflective and Future-Oriented

Question 21 - *Do you think AI can ever be completely unbiased? Justify your answer.*

In my view, AI can never be completely unbiased, though it can be made much fairer. AI learns from data created by people, and that data reflects human history and human bias, so some bias almost always carries through. On top of that, the choices designers make, what data to use and what counts as a fair outcome, are human judgments. What we can do is reduce bias with diverse data, testing, and audits, and stay honest that fairness is something we keep working at rather than a box we tick once.

Question 22 - *How can ethical frameworks ensure responsible innovation in AI?*

Ethical frameworks help by giving developers a shared set of expectations to design against, so ethics is part of the build rather than an afterthought (Slide 25). When frameworks like the OECD Principles or UNESCO's Recommendation are followed, they push teams to embed ethics at every stage, bring in many stakeholders, and keep monitoring systems after launch (Slide 25). They do not guarantee good behavior on their own, but they set a common standard and make it harder to ignore the risks.

Question 23 - *Should governments enforce stricter AI regulations, or should industries self-regulate? Discuss.*

In my view the answer is a mix, leaning toward real government rules for the highest-risk uses. Self-regulation is faster and closer to the technology, but it breaks down when profit and safety pull in different directions, which is what happened before laws like the GDPR and the EU AI Act stepped in (Future of Life Institute, 2024). Government rules set a floor that everyone must meet, while industry standards and self-checks add flexibility on top. The strongest approach uses both: clear laws for serious risks, and responsible practices from companies for everything else.

Question 24 - *What ethical challenges might Quantum AI pose in the future?*

Quantum AI could make systems far more powerful, which raises the stakes on every existing concern (Slide 21). One worry is security, since quantum computing could break much of today's encryption and put private and financial data at risk. Another is that more powerful, faster models could be even harder to explain and control, deepening the black box problem, and could concentrate that power in the few organizations able to afford it (Slide 8, Slide 21). The pace could also outrun our ability to regulate it, which is why ethics needs to be part of the conversation early.

Question 25 - *How might Generative AI reshape intellectual property rights?*

Generative AI unsettles the basic idea of who owns creative work (Slide 19). It is trained on huge amounts of existing text, images, and music, which raises the question of whether using that material without permission or payment infringes the original creators' rights. It also raises the question of who, if anyone, owns what the AI produces, the user, the company, or no one (Slide 19). This is pushing courts and lawmakers to rethink copyright and authorship for work that a machine helped make (Slide 19).

Question 26 - *What are the moral responsibilities of AI developers toward end-users?*

Developers owe users systems that are safe, fair, and honest. That means testing for bias, being transparent about what a system can and cannot do, protecting user data, and not deploying something they know could cause harm (Slide 12, Slide 26). It also means keeping a human path for appeal when the AI gets something wrong, and taking responsibility for outcomes rather than hiding behind the technology (Slide 9). In short, developers should treat the people affected by their systems as the priority, not an afterthought.

Question 27 - *How can AI sustainability contribute to global ethical goals?*

Sustainable AI means building and running systems in ways that do not waste energy or harm the environment, which the deck names as a priority for the next generation of AI (Slide 26). Large models use a great deal of power, so designing them to be efficient supports climate goals and the wider aim of AI benefiting the planet, not just profit (Slide 26). Used well, AI can also advance global goals directly, for example by improving healthcare, education, and resource use, as long as the benefits are shared fairly (Slide 4).

Question 28 - *What future jobs might emerge to address ethical issues in AI systems?*

As AI spreads, new roles are appearing to keep it responsible. These include AI ethicists who guide ethical design, AI auditors who test systems for bias and safety, and AI governance or compliance officers who make sure systems meet laws like the EU AI Act (Slide 23, Future of Life Institute, 2024). There is also growing demand for people who can explain AI to the public and retrain workers whose jobs change, which fits the deck's point about reskilling and upskilling (Slide 23).

Question 29 - *In your opinion, what will be the most pressing AI ethics issue in 2035, and why?*

In my opinion, the most pressing issue by 2035 will be misinformation and the loss of trust in what is real. As generative AI keeps improving, fake video, audio, and text will be almost impossible to tell from the real thing, which can distort elections, damage reputations, and make people doubt genuine evidence. Bias and privacy will still matter, but the sheer scale and believability of AI-generated content is what worries me most, because a society that cannot agree on what is true struggles to make good decisions. This is why I think authenticity and content verification will be central by then.

Question 30 - *Reflect on your role as a future professional. How would you ensure ethical practices in AI development?*

As a future professional in AI and technology, I would strive to check data and models for bias before anything ships, be honest about what a system can and cannot do, and protect the privacy of the people whose data I use. My own path is in AI and robotics for the energy industry, where these choices carry real weight. The data I work with, such as subsurface and well-log measurements, feeds expensive and safety-critical decisions, so I would treat its accuracy and honest labeling as an ethical duty and not just a technical one. And because I also work with physical systems like industrial robots, keeping a human in control of automated decisions matters even more, since a wrong call there can put equipment or people at real physical risk. I would also speak up if I saw a system likely to cause harm, even under pressure to move fast. Most of all, I would keep learning as the field and its rules change, so the work I help build stays aligned with human values.

References

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